

Certified Kubernetes Network Engineer (CKNE) Exam Curriculum

A Cloud Native Computing Foundation (CNCF) Publication

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This document provides the curriculum outline of the Knowledge, Skills and Abilities that a Certified Kubernetes Network Engineer (CKNE) can be expected to demonstrate.

CKNE Curriculum

15% - Core Infrastructure and CNI

- Installing and Configuring CNI Plugins
- Managing IPAM and Pod CIDR Allocation
- Using Linux Tools (iptables, ip, tcpdump) for Packet-level Issues
- Troubleshooting Pod Connectivity (DNS, pod-to-pod)
- Configuring Multi-interface Pods

20% - Advanced Traffic Management

- Optimizing LLM Traffic
- Implementing Routing to Expose Networks
- Configuring Egress Gateways for Cluster Exit Traffic
- Implementing Cross Cluster Service Discovery and Load Balancing

25% - Network Security and Policy

- Securing Traffic with Network Policies
- Implementing Node and Pod Level Encryption
- Managing TLS Certificates for Gateway API
- Implementing Pod-level Authentication and Authorization

25% - Service Networking and DNS

- Configuring L4 Services
- Understanding kube-proxy and CNI Alternatives
- Customizing coreDNS for Services
- Troubleshooting Service Network Traffic
- Configuring Pod Endpoint Availability
- Managing Traffic with the Gateway API (Gateway, HTTPRoutes)

15% - Observability

- Analyzing Network Health Using Metrics
- Troubleshooting End-to-End Network Performance with Tracing
- Auditing Traffic with Logs



kubernetes



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Cloud native computing uses an open source software stack to deploy applications as microservices, packaging each part into its own container, and dynamically orchestrating those containers to optimize resource utilization. The Cloud Native Computing Foundation (CNCF) hosts critical components of those software stacks including Kubernetes, Fluentd, Linkerd, Prometheus, OpenTracing and gRPC; brings together the industry's top developers, end users, and vendors; and serves as a neutral home for collaboration. CNCF is part of The Linux Foundation, a nonprofit organization. For more information about CNCF, please visit: <https://cncf.io/>.